score' = {1..100}
score = {0..100}
money = {1..1000}

s : score
m : money

s = m

a) statycznie OK, dynamicznie MOŻE → m&score ✓ / → m&score fail

b) statycznie NIE

b') s' : score'
s = s' OK
(podtypowanie)

c) implicit cast
X = y ⇝ X = cast_X(y)

s = 0
s = 500
s = 1

xs = {0..5}
ys = {6..10}
x : xs
y : ys
x = y

HL

fire : board -> int (p) =
if p = v17 then 10 else
if p = v24 then 8 else
!

s = cast(m)

cast =

fire(square) ?

is square in traps:
fire(cast to trap(square))

X → y
! id : x ∈ X ⇒ y
⊥ other

I m = 0 II s = 1
m = s
n = (money)s

{? up(*pos) == !}

up(up(up(... up(*pos) ...))) == !

[X = !]

{? up^8(n) == !}

LL

fire[v17] = 10
fire[v24] = 8

class X {}
class Y : X {}
func f(x : X) {..}
f(new Y) OK Y ⊆ X

1 !(null) jest; dla nieznanych wartości dostajemy ! ; każda domena ma !

2 ! (null) jest definiowany i jest zwykłym symbolem

1 czy ¬1

→ 1
¬ !
¬
¬
¬
¬

score' = {1..100}
Score = {0..100}
money = {1..1000}

s : score
m : money

s = m | s = 500, s = 1

a) statycznie OK, dynamicznie MOŻE → m ≤ score, → m ≤ score fail

b) statycznie NIE

b') s' : score'
s = s' OK
(podtypowanie)

c) implicit cast
X = y ⇝ X = cast(y)

xs = {0..5}
ys = {6..10}
x : xs
y : ys
x = y

HL

fire : board → int (p) =
if p = v17 then 10 else
if p = v24 then 8 else
!

s = const 'm'
cast =

x = 1
y = x
x = 2

fire(square) ?

is square in traps:
fire(square to trap(square))

X → Y
! id : x ∈ X → y
⊥ other

I m = 0 II s = 1
m = s
n = (money)s

{? up(*pos) == !}

up(up(up(... up(*pos) ...))) == !

[X = !]

{? up^8(n) == !}

up : pos → pos
val : pos → score
val(up^n(p))

[x = !], źle
[x[k] = !], OK, usuń k (delete)
[x[k] = y], OK, zmień lub dodaj

- g : set → set [default = x] ← x ∈ set, value of g, not its reference
  f : set → (set → set) [default = g]
- nie ma ! (null'i)
- domeny są rozłączne